Artificial Intelligence and Machine Learning – Study Guide

Sample PDF for Week 13 Retrieval-Augmented Generation (RAG)

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of creating computer systems that perform tasks commonly associated with human intelligence. Examples include understanding language, recognizing patterns, recommending content, and supporting decisions. AI systems are designed for specific goals and depend on data, algorithms, and evaluation.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

2. Machine Learning

Machine Learning (ML) is a part of AI. Instead of writing every rule by hand, developers train a model using examples. Supervised learning uses labelled examples, unsupervised learning searches for patterns in unlabelled data, and reinforcement learning learns through feedback from actions.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

3. Data Preparation

A machine learning project begins with understanding the data. Preparation can include removing duplicates, correcting inconsistent formats, handling missing values, and selecting useful features. Training and test data should be separated so that evaluation measures performance on examples the model did not learn from.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

4. Neural Networks

A neural network contains connected layers that transform input values. During training, the model compares predictions with expected outputs and adjusts parameters to reduce error. Neural networks are used for image recognition, speech processing, and language tasks. Their performance depends on data, model design, and training.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

5. Natural Language Processing

Natural Language Processing (NLP) enables computers to process human language. Common tasks include tokenization, translation, summarization, sentiment analysis, and question answering. Text can be represented numerically so that algorithms can process patterns and relationships between words and sentences.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

6. Large Language Models

Large Language Models (LLMs) learn statistical patterns from large text collections. They generate responses based on the input prompt and the context available to them. Their responses can be fluent but may contain errors. Users should verify important information and avoid sharing confidential data.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

7. Text Embeddings

An embedding is a list of numbers that represents information such as a sentence or document. Embedding models map text with related meaning to nearby regions of a vector space. Similarity search compares a query embedding with stored document embeddings to find relevant content.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

8. Vector Databases and ChromaDB

A vector database stores embeddings together with document identifiers and metadata. ChromaDB supports adding documents and searching for similar vectors. Metadata such as title, source, and category helps users understand where a retrieved passage came from.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

9. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines document retrieval with language generation. First, a question is converted into an embedding. The system retrieves relevant text chunks from a vector database. Those chunks are supplied to a language model as context, and the model produces an answer grounded in the retrieved information.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

10. Document Chunking

Long documents are split into smaller chunks before embedding. Chunk size controls how much text is stored in each vector, while overlap preserves some context between neighboring chunks. Very small chunks may lose context; very large chunks may contain unrelated information. Testing different settings helps choose a useful balance.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

11. Evaluating a RAG System

A RAG system should be evaluated for retrieval relevance, answer correctness, and whether the answer is supported by the retrieved context. Test questions should cover different sections of the source document. A system should say when the document does not contain enough information instead of inventing an answer.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.

12. Responsible Use of AI

Responsible AI use includes privacy, fairness, transparency, accessibility, and human oversight. Users should understand the limitations of generated answers. In education and professional settings, AI should support learning and informed decisions rather than replace careful judgement.

Key idea: Understand the concept, check the evidence, and evaluate the system before relying on its output.